

NHANES-Report_Week4_Exercise4

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Overview & Dataset Introduction

The datasets were obtained from the **National Health and Nutrition Examination Survey (NHANES)** <https://www.cdc.gov/nchs/nhanes/>. NHANES is a program of studies designed to assess the health and nutritional status of adults and children in the United States (**Centers for Disease Control and Prevention (CDC) 2013-2014; 2015-2016; 2017-2018**). The resulting data are unique in that they combine interviews with physical examinations and laboratory data. In this report, we present a descriptive examination of a cleaned, multi-wave NHANES dataset (**Demographic Data and Blood Pressure Data** (from Examination Data component)), across three NHANES survey cycles.

Data Summary and Demographic Distribution

The dataset comprises 29,400 total observations spanning three survey waves: 2015 ($N = 10,175$), 2016 ($N = 9,970$), and 2017 ($N = 9,254$).

Across all three waves, the gender proportion remains consistently balanced at approximately 49.1% to 49.2% male participants. Ethnic diversity is recorded using standard NHANES race/ethnicity categorizations, with White, Black, Mexican, Asian, Other and Other Hispanic populations represented.

```
# Import the cleaned dataset
nhanes_data <- import(here("data", "cleaned_NHANES.csv"))

# Compute overall size and male counts per wave
sample_info <- nhanes_data %>%
  group_by(wave_id) %>%
  summarise(
    Total_N = n(),
```

```

Male_N = sum(gender == "Male", na.rm = TRUE),
Male_Pct = (Male_N / Total_N) * 100,
.groups = "drop"
)

# Demographic breakdown by ethnicity_2
ethnicity_info <- nhanes_data %>%
  group_by(wave_id, ethnicity_2) %>%
  summarise(
    Count = n(),
    .groups = "drop"
  ) %>%
  left_join(sample_info %>% select(wave_id, Total_N), by = "wave_id") %>%
  mutate(
    Pct = (Count / Total_N) * 100,
    Formatted_Ethnicity = paste0(format(Count, big.mark = ","),
                                   " (", sprintf("%.1f", Pct), "%)")
  ) %>%
  select(wave_id, ethnicity_2, Formatted_Ethnicity) %>%
  pivot_wider(names_from = ethnicity_2, values_from = Formatted_Ethnicity)

# Merge overall sample & gender info with ethnicity table
demo_table <- sample_info %>%
  mutate(
    `Total Sample Size (N)` = format(Total_N, big.mark = ","),
    `Male Count (%)` = paste0(format(Male_N, big.mark = ","),
                              " (", sprintf("%.1f", Male_Pct), "%)")
  ) %>%
  select(wave_id, `Total Sample Size (N)`, `Male Count (%)`) %>%
  left_join(ethnicity_info, by = "wave_id") %>%
  rename(Wave = wave_id)

# Render formatted LaTeX table
demo_table %>%
  kable(
    format = "latex",
    booktabs = TRUE,
    caption = "NHANES Sample Size, Gender, and Ethnicity Distribution Across Survey Waves",
    col.names = c("Wave", "Total N", "Male N (%)",
                  "Asian", "Black", "Mexican",
                  "Other", "Other Hispanic", "White"),
    align = "c"
  ) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down")) %>%
  add_header_above(c(
    " " = 1,
    "Sample Overview" = 2,
    "Ethnicity Breakdown (N, %)" = 6
  )) %>%
  column_spec(1, bold = TRUE) %>%
  column_spec(2, bold = TRUE, color = "#1A365D") %>%
  row_spec(0, bold = TRUE) %>%
  footnote(general = "Percentages are calculated relative to the total wave sample size.")

```

Table 1: NHANES Sample Size, Gender, and Ethnicity Distribution Across Survey Waves

Wave	Sample Overview		Ethnicity Breakdown (N, %)					
	Total N	Male N (%)	Asian	Black	Mexican	Other	Other Hispanic	White
2015	10,175	5,003 (49.2%)	1,074 (10.6%)	2,267 (22.3%)	1,730 (17.0%)	470 (4.6%)	960 (9.4%)	3,674 (36.1%)
2016	9,971	4,892 (49.1%)	1,042 (10.5%)	2,129 (21.4%)	1,921 (19.3%)	505 (5.1%)	1,308 (13.1%)	3,066 (30.7%)
2017	9,254	4,557 (49.2%)	1,168 (12.6%)	2,115 (22.9%)	1,367 (14.8%)	634 (6.9%)	820 (8.9%)	3,150 (34.0%)

Note:

Percentages are calculated relative to the total wave sample size.

Key highlights from the demographic summary in Table 1 include:

- **Sample Size:** Wave 2015 contained the largest cohort ($N = 10,175$), followed by Wave 2016 ($N = 9,970$) and Wave 2017 ($N = 9,254$).
- **Sex Ratio:** The male representation was nearly identical across waves: 49.2% in 2015, 49.1% in 2016, and 49.2% in 2017.
- **Ethnic Composition:** White participants comprised the largest single ethnic group (30.8% to 36.1%), followed by Black participants (21.4% to 22.9%), Mexican American (14.8% to 19.3%), Asian (10.5% to 12.6%), and Other Hispanic (8.9% to 13.1%).

Systolic Blood Pressure Analysis

Systolic blood pressure (SBP) readings were measured up to four times during individual examination sessions. To establish a reliable metric, an **average systolic blood pressure** was calculated for each individual by averaging all non-missing SBP values across readings 1 through 4.

```
# Calculate average SBP per participant
```

```
nhanes_sbp <- nhanes_data %>%
  rowwise() %>%
  mutate(
    avg_sbp = mean(c_across(c(
      systolic_bp_1, systolic_bp_2,
      systolic_bp_3, systolic_bp_4
    )), na.rm = TRUE)
  ) %>%
  ungroup() %>%
  filter(!is.nan(avg_sbp))
```

```
# Compute summary statistics by wave
```

```
sbp_stats <- nhanes_sbp %>%
  group_by(wave_id) %>%
  summarise(
    N_Valid = n(),
    Min_SBP = min(avg_sbp),
    Max_SBP = max(avg_sbp),
    Median_SBP = median(avg_sbp),
    Mean_SBP = mean(avg_sbp),
    SD_SBP = sd(avg_sbp),
    .groups = "drop"
  ) %>%
  mutate(
```

Table 2: Summary Statistics of Average Systolic Blood Pressure (mmHg) by Wave

Wave	Valid N	Systolic Blood Pressure Metrics (mmHg)				
		Min	Max	Median	Mean	Std Dev
2015	7,528	64.7	228.7	115.3	118.3	18.1
2016	7,363	74.0	231.3	117.3	120.4	18.4
2017	6,713	72.7	234.0	118.0	121.7	20.3

Note:

Average SBP is calculated across all available individual readings. SBP values are measured in mmHg.

```

N_Valid = format(N_Valid, big.mark = ","),
Min_SBP = sprintf("%.1f", Min_SBP),
Max_SBP = sprintf("%.1f", Max_SBP),
Median_SBP = sprintf("%.1f", Median_SBP),
Mean_SBP = sprintf("%.1f", Mean_SBP),
SD_SBP = sprintf("%.1f", SD_SBP)
)

# Render formatted SBP table
sbp_stats %>%
  kable(
    format = "latex",
    booktabs = TRUE,
    caption = "Summary Statistics of Average Systolic Blood Pressure (mmHg) by Wave",
    col.names = c("Wave", "Valid N", "Min", "Max", "Median", "Mean", "Std Dev"),
    align = "c"
  ) %>%
  kable_styling(latex_options = c("striped"),
    full_width = FALSE) %>%
  add_header_above(c(" " = 2, "Systolic Blood Pressure Metrics (mmHg)" = 5)) %>%
  column_spec(1, bold = TRUE) %>%
  column_spec(6, bold = TRUE, color = "#2B6CB0", background = "#EDF2F7") %>%
  row_spec(0, bold = TRUE) %>%
  footnote(
    general = "Average SBP is calculated across all available individual readings. SBP values are measu
  )

```

As detailed in Table 2, valid blood pressure measurements were available for 21,604 total participants (7528 in 2015, 7363 in 2016, and 6713 in 2017). Analysis of average SBP reveals:

1. **Mean and Median Trends:** Mean average SBP rose slightly across survey waves, increasing from 118.3 mmHg in Wave 2015 to 120.4 mmHg in Wave 2016, and 121.7 mmHg in Wave 2017. Median SBP demonstrated a parallel trend (115.3 mmHg, 117.3 mmHg, and 118.0 mmHg, respectively).
2. **Variability:** Standard deviations remained stable across 2015 (18.1 mmHg) and 2016 (18.4 mmHg), with a slight increase in 2017 (20.3 mmHg).
3. **Range:** Minimum recorded averages ranged from 64.7 to 74.0 mmHg, while maximum recorded averages ranged from 228.7 to 234.0 mmHg.

References

Centers for Disease Control and Prevention (CDC). 2013-2014; 2015-2016; 2017-2018. *National Health and Nutrition Examination Survey (NHANES) Data*. 2013-2014; 2015-2016; 2017-2018. <https://www.cdc.gov/nchs/nhanes/index.htm>.